WHITEBOX: Mathematical Foundations

Gauge Symmetry, Invariance, and the Structure of Explanations in Neural Networks

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Abstract. We set out the mathematics underlying a program to audit mechanistic interpretability. A feedforward network's parameters carry a gauge group $G = \prod_\ell S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}$ acting on the fibers of the realization map; a quantity measured inside the network is well-defined only if it is a function on the quotient. We formalize the invariance criterion (the mechanistic extension of the Implementation Invariance axiom), classify candidate quantities as invariant, covariant, or artifact, and derive the scaling covariance that invalidated an earlier persistence-based statistic. We show that explanation multiplicity within a single generic network is never gauge (the stabilizer is trivial) and reformulate uniqueness as π_0 of the valid-circuit locus modulo G across independent trainings. We give the Čech construction measuring the obstruction to gluing local explanations, the Marchenko–Pastur null for decomposition residue, and the parameterized-complexity framing of explainability. Four questions are posed for a topologist: completeness of G, MSO-expressibility of circuit discovery, generators of the invariant ring $\mathbb{R}[\Theta]^G$, and uniqueness up to gauge.

1 Setting

A feedforward network with architecture $\mathcal{A}$ is a family of functions $f_\theta : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ indexed by parameters $\theta \in \Theta \cong \mathbb{R}^P$. For a ReLU multilayer perceptron with layer widths $n_0, \ldots, n_L$,

$$f_\theta(x) = W_L\, \sigma(W_{L-1} \cdots \sigma(W_1 x + b_1) \cdots + b_{L-1}) + b_L, \qquad \sigma(t) = \max(t, 0) \text{ coordinatewise.}$$

Fix a finite input set X (for Boolean tasks, the full truth table). The *realization map* is

$$R : \Theta \to \mathbb{R}^{|X| \cdot d_{\text{out}}}, \qquad R(\theta) = (f_\theta(x))_{x \in X}.$$

Training finds θ with $R(\theta)$ equal (or close) to a target. The *fiber* $R^{-1}(R(\theta))$ is the set of parameters computing the same function on X.

Throughout, an *explanation* of a behavior is a structured object attached to θ: a sub-circuit (a subset of hidden units, with the complement ablated), possibly with an *interpretation* (an assignment of Boolean gates to units consistent with their activations). The interpretability literature's problem is that many explanations are valid for one θ and there is no principled way to choose among them.

2 The gauge group

Definition 2.1 (Gauge group)**.** The gauge group $G = G_{\mathcal{A}}$ is the group of maps $g : \Theta \to \Theta$ with $R \circ g = R$ on all of Θ (functional equivalence for every parameter, not just a particular fiber).

For ReLU networks, G contains:

1. **Permutations.** For each hidden layer ℓ, $\pi_\ell \in S_{n_\ell}$ acting by $W_\ell \mapsto P_{\pi_\ell} W_\ell$, $b_\ell \mapsto P_{\pi_\ell} b_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} P_{\pi_\ell}^\top$.
2. **Positive scalings.** For each hidden unit u in layer ℓ and $\alpha_u > 0$: multiply row u of W_ℓ and $b_{\ell,u}$ by α_u, divide column u of $W_{\ell+1}$ by α_u. This uses positive homogeneity $\sigma(\alpha t) = \alpha\sigma(t)$.

So $G \supseteq \prod_\ell \left(S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}\right)$, a product of hyperoctahedral-type groups (permutations acting on scaling tori). The scaling part is a connected Lie group of dimension $\sum_\ell n_\ell$.

Definition 2.2 (Functional dimension and hidden symmetries)**.** At θ, the fiber dimension is $\dim\ker dR_\theta = P - \operatorname{rank} dR_\theta$ (generically). Grigsby et al. call $\operatorname{rank} dR_\theta$ the *functional dimension*. Any excess of $\dim\ker dR_\theta$ over the dimension of the known-symmetry orbit through θ certifies *hidden symmetries* — directions in the fiber not generated by permutations and scalings.

What we compute (test F3). For each trained θ: $\operatorname{corank}(dR_\theta)$ numerically, compared to $\sum_\ell n_\ell$ (scaling dimension; permutations are discrete). Known sources of excess: dead units (zero activation on all of X, contributing free directions), duplicated units (identical incoming weights), and units whose pre-activations never change sign on X (locally linear, so the ReLU is invisible). Each is detected and classified; unexplained excess is logged.

Question 2.1 (Completeness — for JVHM)**.** For which classes of ReLU architectures and generic θ is $\ker dR_\theta$ exactly the tangent space to the permutation–scaling orbit? Grigsby–Lindsey–Rolnick have partial results (functional dimension bounds; "hidden" positive-scale symmetries); a clean characterization of when the known group is the whole stabilizer of the fiber would settle the foundation of everything below.

3 The invariance criterion

Definition 3.1 (Invariance classes)**.** Let $Q : \Theta \to \mathbb{R}^m$ be any quantity someone measures about a network. Sample $g_1, \ldots, g_n \in G$ (random permutations composed with random positive scalings, $\log\alpha \sim \mathrm{Unif}[\log 0.1, \log 10]$). Compute the per-component coefficient of variation of $\{Q(g_i\theta)\}$. Classify Q as

- *invariant* if $Q(g\theta) = Q(\theta)$ for all g (CV below numerical tolerance);
- *covariant* if $Q(g\theta) = \rho(g)\, Q(\theta)$ for a known character or representation ρ — e.g. Q scales as α^k under a single-unit scaling (we fit the exponent k);
- *artifact* otherwise.

The criterion. An interpretability claim about the internals of θ is well-defined iff it is a function on Θ/G, i.e. invariant. Covariant quantities become invariant after normalization by the character. Artifacts describe coordinates, not the network. This is the mechanistic extension of the *Implementation Invariance* axiom of Sundararajan–Taly–Yan (2017), which the attribution literature states for input-level explanations of functionally equivalent networks and has never applied to internal quantities.

Expected classification (test F10), with proofs where trivial:

- Products $W_L W_{L-1} \cdots W_1$ along paths with fixed activation pattern: invariant (permutations conjugate, scalings cancel telescopically).
- Singular values of a single W_ℓ: permutation-invariant (conjugation by permutation matrices), *not* scaling-invariant.

- Persistence diagrams of the activation cloud $A_\ell(\theta) = \{h_\ell(x;\theta) : x \in X\} \subset \mathbb{R}^{n_\ell}$: permutation-invariant because the Vietoris–Rips construction depends only on pairwise Euclidean distances, and P_π is an isometry. Under a *uniform* scaling α of layer ℓ, all distances scale by α, every birth/death scales by α, total persistence $\sum(d_i - b_i)^2$ scales by α^2: covariant of degree 2. Under *non-uniform* per-unit scalings the map is a diagonal linear map, not an isometry, and persistence is neither invariant nor covariant — this is measured, not assumed.
- Per-unit activation magnitudes, per-unit attribution scores, top-k unit sets, single-unit weight norms: artifacts (scaling changes them arbitrarily; permutation relabels them).
- Single-direction "features" (as in dictionary methods): the direction transforms as a vector under P_π and under the diagonal scaling; comparable across gauges only after alignment (Procrustes / permutation matching). We test both raw and aligned.

Remark 3.1 (The v1 error, stated precisely)**.** Version 1 used total persistence $TC = \sum_i (d_i - b_i)^2$ as a structural measure. Under a uniform contraction of the activation cloud by λ, $TC \mapsto \lambda^2 TC$. Networks trained on fully random labels never fit and their activations contract; the observed "2,900$\times$ collapse" was the λ^2 factor. The feature-shuffle null (independently permute each coordinate across points, destroying all joint structure while preserving marginals) retained 77–86% of TC, confirming the statistic was measuring dispersion. The corrected statistic (test Z5) normalizes persistence by a scale estimate (e.g. divide all diagram coordinates by the median pairwise distance), yielding a degree-0 — hence invariant — quantity.

4 Orbits, quotients, and the uniqueness question

The natural object is the quotient Θ/G, or better, the quotient of a fiber $F = R^{-1}(y)$ by G. Let $\mathcal{E}(\theta)$ denote the set of valid explanations of θ.

4.1 The within-network tautology (what went wrong two weeks ago)

Two explanations $E, E' \in \mathcal{E}(\theta)$ of the *same* θ are gauge-equivalent iff there is $g \in \mathrm{Stab}_G(\theta)$ with $g \cdot E = E'$. But for generic θ (distinct incoming weight vectors, no dead units), $\mathrm{Stab}_G(\theta)$ is trivial: a permutation that fixes θ must map each unit to a unit with identical weights, and a scaling that fixes θ must be the identity. Hence every $E \in \mathcal{E}(\theta)$ is its own orbit and $K/N = 1$ identically. The experiment could not have returned anything else. The observed $K/N = 1.0$ across 3,600 networks is a theorem, not a finding.

Proposition 4.1 (Two sources of multiplicity)**.** Explanation multiplicity in the sense of Méloux et al. decomposes as: (a) *coordinate multiplicity* — distinct θ, θ' in the same fiber related by G, whose explanations correspond under g; (b) *structural redundancy* — distinct $E, E' \in \mathcal{E}(\theta)$ for a single θ, which are never gauge-related for generic θ. Only (a) is "gauge."

The planted control showed (b) is combinatorial: if θ contains two disjoint sub-circuits each computing the target, then every recombination (part of one, part of the other) is also valid, so duplicating a circuit multiplies $|\mathcal{E}|$ far beyond a factor of two. Redundancy is a Hydra-type phenomenon, not a coordinate one.

4.2 The corrected test: cross-fiber orbits

Train $\theta_1, \ldots, \theta_s$ from independent seeds to the same target y. Extract minimal circuits $C_j \subset \theta_j$. Define $C_i \sim C_j$ iff there is $g \in G$ with $g \cdot (\theta_i|_{C_i}) \approx \theta_j|_{C_j}$ (weights match on the circuit's support up to tolerance, after

quotienting hidden-symmetry directions). The number of equivalence classes among $\{C_j\}$ is the number of *distinct mechanisms* realized across independent trainings.

Topologically: the valid-circuit locus in the fiber, $\mathcal{C}(F) = \bigsqcup_{\theta \in F} \mathcal{E}(\theta)$, carries a G-action; we are computing $\pi_0\big(\mathcal{C}(F)/G\big)$, sampled by independent trainings. Test Z4 additionally asks whether sampled circuits in one class are joined by paths of valid circuits (path-connectedness of the class, not just orbit membership).

Conjecture 4.1 (Uniqueness up to gauge)**.** For a suitable class of Boolean targets and architectures, and a suitable notion of minimal circuit, $\pi_0(\mathcal{C}(F)/G)$ is a single point: the mechanism is unique up to symmetry.

We expect this to be false in general (distinct named algorithms exist for XOR) and true for some targets; the count is the result either way. Adjacent literature: "convergent learning" (Li et al. 2015) and permutation-alignment / re-basin work, which align *whole networks* across seeds by permutation; we align *circuits* and include scalings and hidden symmetries. Novelty check required before write-up.

5 Persistent homology — what is used and why it survives

Definition 5.1. For a finite point cloud $A \subset \mathbb{R}^n$ and $\epsilon \geq 0$, the Vietoris–Rips complex $\mathrm{VR}_\epsilon(A)$ has a simplex on $\{a_0, \ldots, a_k\}$ iff $\|a_i - a_j\| \leq \epsilon$ for all i, j. The filtration $\epsilon \mapsto \mathrm{VR}_\epsilon(A)$ induces persistence modules $H_k(\mathrm{VR}_\bullet(A))$; the persistence diagram $D_k(A)$ records birth–death pairs (b_i, d_i) of its indecomposable summands. Distances between diagrams: $W_p(D, D') = \inf_\gamma \left(\sum \|x - \gamma(x)\|^p \right)^{1/p}$ over partial matchings with unmatched points sent to the diagonal.

Properties used. (i) $D_k(A)$ depends only on the distance matrix of A, hence is invariant under isometries of $\mathbb{R}^n$ — in particular under coordinate permutations. (ii) Under $A \mapsto \lambda A$, $D_k \mapsto \lambda D_k$ pointwise. (iii) Stability: $W_\infty(D_k(A), D_k(A')) \leq d_H(A, A')$ (Cohen-Steiner–Edelsbrunner–Harer), so small perturbations of activations give small diagram changes.

Where it appears. As one row of the invariance audit (F10), with its scaling covariance measured; as the null-tested topological summary of the dark-matter residue (K3); as the corrected, normalized statistic (Z5).

5.1 Zigzag and multiparameter persistence (tests Z1, Z2)

Version 1 computed a separate diagram at each training checkpoint and compared consecutive diagrams by W_2 ("structural velocity"). This cannot track *which* feature persists across time. Zigzag persistence (Carlsson–de Silva) handles a sequence of spaces with maps in alternating directions,

$$A_0 \hookleftarrow A_0 \cup A_1 \hookrightarrow A_1 \hookleftarrow A_1 \cup A_2 \hookrightarrow \cdots,$$

built here from consecutive checkpoint clouds; its decomposition gives intervals in *training time* for individual homology classes. Two-parameter persistence (scale $\times$ training fraction) has no complete discrete invariant; we use the rank invariant / Hilbert function (Carlsson–Zomorodian; RIVET) and test whether it predicts circuit-discovery cost better than one-parameter summaries.

5.2 Graph persistent homology (test Z3)

For an attribution graph Γ (nodes = units/heads, edge weights = attribution strength), the clique complex filtered by decreasing edge weight gives H_1 intervals measuring cycle structure in the effective connectivity. Used as a candidate legibility feature.

6 The gluing problem: a Čech computation

Fix a behavior (a task) and a set of inputs $\mathcal{X}$. Cover $\mathcal{X}$ by overlapping regions $\{U_i\}$ (soft clusters in embedding space). On each U_i, circuit discovery yields a local explanation E_i (a weighted edge set). On $U_i \cap U_j$, each E_i restricts (re-scored on the overlap prompts) to $E_i|_{ij}$.

This is a presheaf $\mathcal{F}$ of explanations on the nerve $N(\{U_i\})$: sections over U_i are circuits; restriction is re-scoring. Two restricted sections *agree* if their thresholded edge sets have Jaccard similarity $\geq \tau$. Then:

- H^0: the number of connected components of the agreement graph on the nerve's vertices — the number of mutually consistent explanation families (distinct mechanisms).
- H^1: the obstruction to choosing a single global section. Concretely, we compute the cycle space of the disagreement-weighted nerve: a nontrivial 1-cocycle is a cyclic chain of overlaps whose pairwise disagreements cannot be resolved by any global re-assignment of sections.

Attribution. The sheaf-theoretic framing of local-to-global explanation is due to arXiv 2603.17141 (2026) and to the "gauge theory of superposition" sheaf-atlas paper; we cite both. Our contribution is the first concrete computation on a real model with an honest sanity gate: a task known to use a single mechanism must return $H^1 \approx 0$, or the construction is rejected.

Remark 6.1. The choice of agreement relation makes this a *thresholded* presheaf; varying τ produces a filtration of nerves, and one could compute persistent H^1 across τ — a natural refinement we may adopt if the single-τ obstruction is unstable.

7 Random matrix theory for the residue

Let $e_t = x_t - \mathrm{SAE}(x_t) \in \mathbb{R}^d$ be reconstruction residuals over n tokens, $\Sigma = \frac{1}{n}\sum_t e_t e_t^\top$. If the e_t were i.i.d. isotropic noise of variance σ^2, the empirical spectral distribution of Σ converges (as $d, n \to \infty$, $d/n \to \gamma$) to the Marchenko–Pastur law with support $[\sigma^2(1-\sqrt{\gamma})^2,\ \sigma^2(1+\sqrt{\gamma})^2]$, and the largest eigenvalue fluctuates about the upper edge on the Tracy–Widom scale $n^{-2/3}$. Under a finite-rank "spiked" perturbation (signal plus noise), eigenvalues above the BBP threshold separate from the bulk.

What we compute (K2). $\hat{\sigma}^2$ from the bulk median; the number of eigenvalues exceeding the Tracy–Widom $p < 0.01$ edge; the *structured fraction* $\sum_{\lambda_i > \lambda_+} \lambda_i / \operatorname{tr}\Sigma$. Nulls: synthetic Gaussian residue with matched marginals (must give ≤ 1 outlier), and the real residue with coordinates independently shuffled across tokens.

Caveats stated in advance. Residuals are not i.i.d. across tokens (context dependence) and not isotropic; the MP bulk is therefore a conservative null for "no structure beyond marginal variances," not a claim about the true noise process. Test K9 (cross-token correlation) addresses the dependence directly.

8 Parameterized complexity and the legibility parameter

Adolfi et al. (2025) prove that natural formulations of circuit discovery (find a minimal sub-circuit reproducing behavior to tolerance) are NP-hard, inapproximable, and that the *global* version (faithful on all inputs) is Σ_2^p-complete.

Definition 8.1. A parameterized problem with input size n and parameter k is fixed-parameter tractable (FPT) if solvable in time $f(k) \cdot n^{O(1)}$.

Many NP-hard graph problems are FPT in treewidth (Courcelle's theorem for MSO-definable properties). Our candidate parameters for the attribution graph Γ_θ are: modularity Q (Newman–Girvan, optimized by Leiden), treewidth $\mathrm{tw}(\Gamma_\theta)$ (upper-bounded by min-fill), and sparsity.

Conjecture 8.1 (Legibility)**.** Circuit discovery restricted to networks whose thresholded attribution graph has treewidth $\leq k$ (equivalently, whose module-interaction graph has bounded width) is FPT in k. Empirically, gradient-trained networks concentrate at small k.

What we measure (V2–V4). Across ≥ 60 trained models spanning regularization regimes: $(Q, \mathrm{tw}, \text{sparsity})$ against the wall time and success of a fixed discovery heuristic. The pre-registered prediction is a change-point (a knee) rather than a linear relation. Golechha et al. have the qualitative result that training for clusterability shrinks circuits; the quantitative diagram and the FPT statement are the open parts.

Question 8.1 (For JVHM)**.** Is the discovery problem, as formalized by Adolfi et al., MSO-expressible on the attribution graph so that Courcelle applies directly? If not, what is the right width parameter?

9 Invariant theory and the completeness of the dictionary

If meaningful quantities are exactly G-invariant functions on Θ (or on activation space), the natural question is whether the ring of invariants $\mathbb{R}[\Theta]^G$ is finitely generated and what generates it. For the scaling torus $T = (\mathbb{R}_{>0})^N$ acting by characters, invariants are generated by monomials in a lattice (Laurent monomials with zero weight) — explicitly computable. For the permutation part, symmetric-function-type generators (power sums of unit-wise invariants) appear. For the semidirect product acting on weight tensors, we expect a description in terms of path products (telescoping products along input–output paths are scaling-invariant; symmetrizing over layer permutations gives permutation invariance).

Question 9.1 (For JVHM)**.** Give a generating set for $\mathbb{R}[\Theta]^G$ with $G = \prod_\ell S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}$ acting on $(W_1, b_1, \ldots, W_L, b_L)$ as in Section 2. Is the "path-product" algebra (all products $W_L[\cdot, u_{L-1}] \cdots W_1[u_1, \cdot]$ over paths, symmetrized) the full invariant ring, or are there additional invariants?

An answer bounds what our "dictionary" (test F10, module `invariants`) can ever contain, and turns the empirical completeness test (predicting behavior from invariants) into a theorem-backed claim.

10 Statistical discipline

Every structure claim is paired at pre-registration with a null: (i) *feature shuffle* — independently permute each coordinate across points, preserving marginals, destroying joint structure; (ii) *label permutation* for any claim conditioned on labels; (iii) *Marchenko–Pastur* for spectral claims; (iv) *planted structure* — synthetic data with known structure of known size, to calibrate detectors and demonstrate power. Bootstrap confidence intervals throughout. Effect sizes reported alongside p-values. A result is believed only after the listed kill attempts fail; the attempts are reported either way.

11 Summary of the mathematical spine

Question	Mathematics	Test block
Which measurements are real	G-action on Θ; invariance classes; characters	F
Is the mechanism unique	$\pi_0(\mathcal{C}(F)/G)$ via cross-seed circuit alignment	F, Z4
Does one global explanation exist	Čech H^0, H^1 of a presheaf of local circuits	E3
What do decompositions miss	Marchenko–Pastur / Tracy–Widom spectral null	K
Which systems are explainable	Parameterized complexity; treewidth / modularity	V
Corrected topological summaries	Normalized persistence; zigzag; multiparameter	Z
Completeness of the dictionary	Invariant theory of $\prod S_n \ltimes (\mathbb{R}_{>0})^n$	theory

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Three of the open questions above — completeness of G, generators of $\mathbb{R}[\Theta]^G$, and FPT in a width parameter — are the theorems this program is built around. The experiments supply evidence; the proofs would supply the foundation.